Hand Calculation (Uliana)

f1 = 23.019741 and cat = No treatment

Firstly, we split training data by class:

y=0 –> healthy plants (900 plants);

y=1 -> unhealthy plants (300 plants).

Category count for y=0 -> 135 (amount of healthy plants from 'No treatment' category)

Category count for y=1 -> 110 (amount of unhealthy plants from 'No treatment' category)

In addition, we split all data into testing set and training set (25% - testing data, 75% - training data), and we are going to use training data since Gaussian Naïve Bayes learns from training data.

Now, for each class:

1) Compute *mean and std. deviation* for every feature.
2) Compute priors
3) For a new point, calculate every density (likelihoods)
4) Multiply: prior * all likelihoods
5) Pick the largest; divide by the total to get a probability (normalization)

Hand calculation for y=0:

1) Both mean μ and std. deviation σ are calculated by the code.

So, μ = 20.083083 (average f1 for healthy plants) and σ = 1.989512 (how many f1 vary among healthy plants)

2) Prior P(y=0) = $\frac{675\ (75\%\ \text{of}\ 900)}{900}$ = 0.75

3) Gaussian density indicates how well f1 = 23.019741 follows the normal distribution of f1 among healthy plants. For numerical features we use this formula:

$$p(x|C_k) = \frac{1}{\sqrt{2\pi * \sigma_k^2}} * e^{\frac{-(x-\mu_k)^2}{2*\sigma_k^2}}$$

$$p(f1|y=0) = \frac{1}{\sqrt{2\pi * 1.989512^2}} * e^{\frac{-(23.019741-20.083083)^2}{2*1.989512^2}}$$

$$= 0.06746009$$

For categorical feature 'No treatment', we do it differently:

<u>Without smoothing:</u>

p(cat=No treatment | y=0) =$\frac{135}{675}$ = 0.20
Thus, 20% of plants in training data have category 'No treatment'

However, the code uses Laplace smoothing in order to avoid that a categorical likelihood becomes zero if category doesn't appear in training data for a specific class.

<u>With Laplace smoothing:</u>

$$\frac{135+1}{675+4} = 0.20029455$$

So, 0.20029455 is categorical likelihood that a plant from 'No treatment' is healthy.

4) Now, multiply *prior* * *all likelihoods* for y=0:

0.75 * 0.06746009 * 0.20029455 = 0.0101339163 (unnormalized score for y=0)

Hand calculation for y=1:

1) μ = 15.098245 and σ = 1.916490

2) Prior P(y=1) = $\frac{225\ (75\%\ of\ 300)}{900}$ = 0.25

3) For numerical feature:

$$p(f1|y=1) = \frac{1}{\sqrt{2\pi * 1.916490^2}} * e^{\frac{-(23.019741-15.098245)^2}{2*1.916490^2}} = 0.00004060$$

For categorical feature:

Without smoothing:

$\frac{110}{225}$ = 0.4889

With Laplace smoothing:

$\frac{110+1}{225+4}$ = 0.48471616

4) Now, multiply *prior* * *all likelihoods* for y=1:

0.25 * 0.00004060 * 0.48471616 = 0.0000049203 (unnormalized score for y=1)

5) NORMALIZATION

Now do this formula:

$$P(C_k|x) = \frac{P(C_k) * p(x|C_k)}{\sum_j P(C_j) * p(x|C_j)}$$

So, the probability that test plant is healthy is:

P(y=0 | x) = $\frac{0.0101339163}{0.0101339163+0.0000049203}$ = **0.99951471 (winner)**

While for unhealthy plant:

P(y=1 | x) = $\frac{0.0000049203}{0.0101339163+0.0000049203}$ = **0.00048529**

We pick the largest score.

So, since there is almost 100% chance to get a healthy plant, the model predicts that.